The ECBench ECDLP performance frontier

Factor bases, symmetry, F4/F5/F6, and root finding

Evidence synthesis and current-binary replay audit · 2026-10-13 · `run-20261013T2025Z`

Boundary. Generic rho remains the cold one-target, setup-inclusive frontier. Compact-orbit index calculus owns the admitted retained-setup online point at n=61: rho/IC 13.9x, 95% interval [9.8, 19.1], 64 unseen targets on an isolated x86_64 host. The newer N83 dense-root point is 18.528x online but exploratory; its complete IC process is 339.068 s against 1.173 s rho for that point. Solver-stage wins do not change the cold frontier.

1. Scope and evidence class

This document reconciles committed measurements at repository commit `759978250c1ce53d7b87bd8a1c2304781fcc4fc4`. It is not a new cross-method session and creates no bound or challenge verdict. Machine-readable values, source paths, derivations, and classifications are in `frontier-data.json`.

requested line	status	evidence boundary
factor bases	complete for catalog; 7 measured arms	10 sealed bases; four native sessions; operation smokes at L0
symmetry	partial	signed Frobenius is native; torsion-symmetrised u-frame remains a legacy non-ECBench study
F4/F5/F6	committed cells covered	K13/R13/K19/R19; matrix m19 and F6 m31 timeouts retained
native root finder	covered	pair/subtraction ECBench engines; current root-finder rows replay exactly
new N83 root finder	not ECBench-integrated	native independent study; 8 GiB table; shared-host online observation

2. Formal boundary

A performance point is keyed by curve/subgroup, exact targets, cold/online/stage interval, counted unit and unpriced work, amortisation target count, memory and resource envelope, and correctness/replay/isolation class. Ratios are formed only when that key matches apart from method identity. A point is Pareto-optimal only if no admitted point is no worse on every declared axis and better on one.

This separates four quantities that cannot be collapsed: setup-excluded online wall and cold end-to-end cost; word XORs/probes/field operations/GAE; lower bounds and complete counts; exploratory L0 wall and admitted isolated wall.

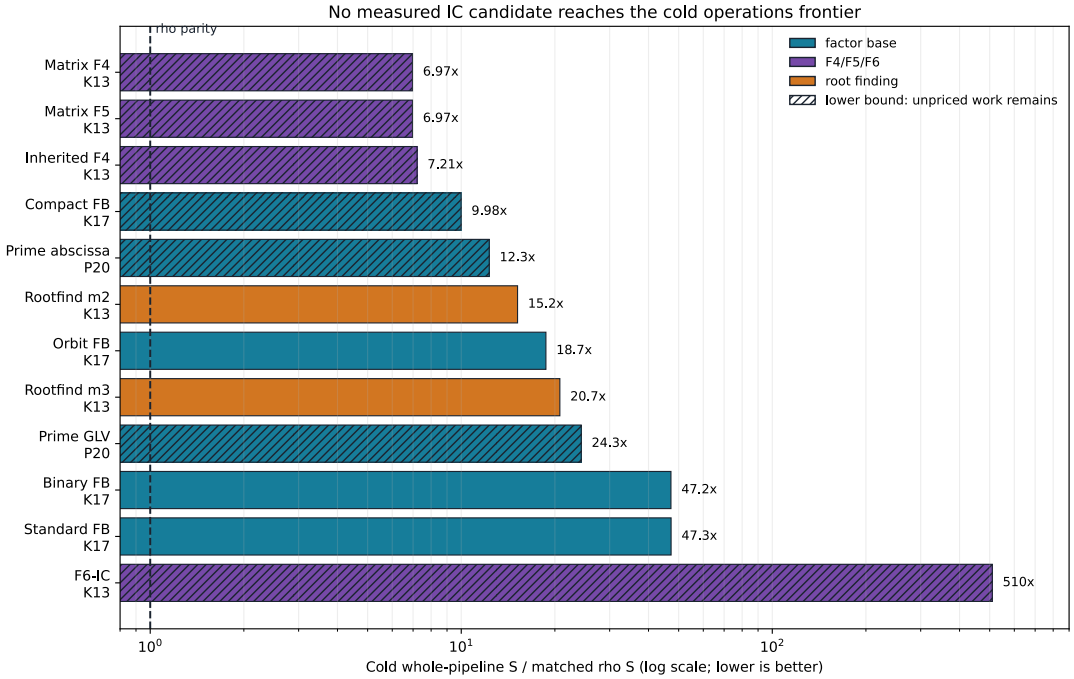
For q targets and reusable preparation B:

$$T_{\mathrm{IC}}(q) = B_{\mathrm{IC}} + qO_{\mathrm{IC}}, \quad T_{\rho(q)} = B_{\rho} + qO_{\rho}$$

At N83, the reported reusable components sum to 339.004 s. With `O_IC=0.063324125 s` and `O_rho=1.17329325 s`, the arithmetic crossover is 305.418, or 306 targets. This is a derived what-if, not a measured batch frontier: it assumes fully reusable setup, constant online cost, no capacity change, and zero rho setup.

3. Canonical cold frontier

axis	incumbent	measurement	coverage
Koblitz medium ops	`rho.signed_frobenius_strong` / `ECBND1h05a2ea918f3c`	1.089x floor [0.927, 1.261]; alpha 0.451 [0.399, 0.512]	144 runs, 6 sizes
Koblitz medium memory	`rho.signed_frobenius` / `ECBND1h46fbc9140ac3`	1.188x floor [0.954, 1.464]; 0.000706 sqrt(r) memory	72 runs, 3 sizes; constant bound
retained-setup online	compact-orbit IC n61 K500	rho/IC 13.9x [9.8, 19.1]	64 unseen points, isolated x86_64
newer online observation	dense root N83 K1798	rho/IC 18.528377	one point, shared host, exploratory



Each bar uses its study's matched generic rho reference and reported estimator; curves, references, and estimators differ, so only parity is shared. Sources and derivations: frontier-data.json. Current audit: ECBHW1h7373da65e433.

Figure 1: Cold whole-pipeline operation gaps to each study's matched generic rho, retaining its reported estimator. Curves, references, and estimators differ, so parity—not bar-to-bar raw cost—is the shared comparison. Hatched bars retain unpriced work and are lower bounds.

Every selected IC bar is above parity. The smallest selected gap is matrix F4/F5 at 6.97x rho-negation on K13; compact orbit is 9.985x its matched signed-Frobenius rho on the n17 factor-base cell. No candidate displaces rho.

4. Factor bases and symmetry

cell	candidate	points / cols	mean S	S/rho	reading
K17 a=1	compact orbit	272 / 8	9.114	9.985	best in matched cell; lower bound
K17 a=1	full orbit	443 / 14	17.081	18.714	complete count
K17 a=1	binary subspace	10 / 5	43.103	47.224	complete count
K17 a=1	standard subspace	11 / 6	43.150	47.275	complete count
K23 a=0	compact orbit	460 / 10	2.996	8.360	separate curve
K23 a=1	cyclotomic w=1	46 / 1	3.427	9.933	separate curve
prime 20	abscissa	64 / 32	11.329	12.300	better than GLV here; lower bound
prime 20	GLV orbit	32 / 16	22.390	24.309	lower bound

The native catalog contains ten sealed bases and reports 10/10 remote verification; seven IC arms have performance rows. Compact orbit is the local Koblitz winner, and ordinary abscissa is the local prime winner. Neither beats rho.

The legacy torsion-symmetrised u-frame has whole-pipeline `S_sym/S_x` 0.84 at n17 and 0.77/0.76 on the two n23 curves, then 4.01 at n31. Its combinatorial controls are 0.89, 0.90, 1.09, and 1.28. The small-size gain is not persistent. The current ECBench registry has no torsion-symmetrised factor-base plug-in.

Signed Frobenius is a separate symmetry: it is native in rho and owns the canonical Koblitz operations frontier. Factor-base orbit folding must not be credited with the walk's automorphism discount.

5. F4, F5, F6-IC

comparison	observation	boundary verdict
matrix F5 / matrix F4	word-XOR ratio 0.6465 K13, 0.6414 R13; same trees	solver-stage engineering win
matrix F4 / inherited F4	198.6x K13, 222.1x R13	inherited F4 is the practical algebraic baseline
F6-IC / inherited F4	solver-XOR ratio about 0.82 at m13 and 0.77-0.78 at m19; relation wall 1.7-1.9x faster	solver-stage win
charged F6-IC S	64.6-68.8x inherited at m13; 1,303-1,353x at m19	relabelling/accounting loss
matrix m19	four 1,800 s timeouts	censored
F6 ladder m31	both arms hit 3,600 s on target 1	protocol decision open

Every IC arm is above matched rho on every target in the cross-family cells. F5 and F6 improve a subroutine, not the cold ECDLP frontier.

6. Root finding

Three root-finder lines are distinct.

1. **Native ECBench pair/subtraction engines.** K13 pair-table root finding has complete S 28.383 (m2) and 38.822 (m3): 15.170x and 20.749x matched signed-Frobenius rho. It matches MITM operations exactly. Descriptive L0 wall favours it over inherited F4, but a complete count cannot be divided by an F4 lower bound to claim an operation speedup.
2. **PDP root-finding study.** The table/enumeration oracles beat F4 within the decomposition stage, but every measured slope equals its generic pair-table or enumeration floor. Classification: engineering, not an exponent change.
3. **N83 dense five-byte index.** K1798 lowers mean probes from 300,021.90625 to 169,069.8125. The ratio is 0.563525; the paired reduction is 130,952.09375 [52,057.703125, 209,990.828125]. It uses an 8 GiB root table and is not registered in `ecbench methods`.

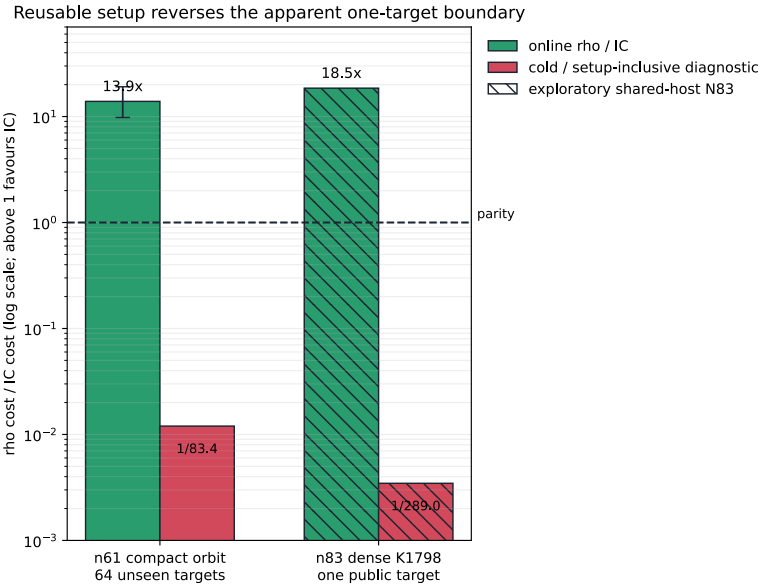


Figure 2: The same methods cross parity only after reusable preparation is excluded. Both series use rho cost / IC cost. The n61 online interval and cold ratio are reported measurements. N83 is a one-point shared-host observation; its lower bar divides rho online by the complete IC process and is a setup diagnostic, not a matched cold ratio.

7. Current-binary replay audit

Build: Rust 1.94.1, current commit, auditor SHA-256 `53654ef1c0e93ed02025acd8188514d2674c4da2f57f25f61db40a510a422950`, hardware class `ECBHW1h7373da65e433`.

session	audit	receipt prefix	reading
factor bases K17	8/8 identical	`9306d33e…`	current-binary pass
F4/F5/F6 K13	7/7 identical	`0304c496…`	representative pass covering all six arms; not replay-all
rootfinder K13 sample	8/8 identical	`4814136e…`	sampler selected rho only
rootfinder K13 replay-all	120/168 identical; 48 failures	`6110727a…`	all rootfinder/rho rows pass; every old inherited-F4 row differs in solver phase counts

The replay-all failure is preserved. It changes no scalar, factor base, trial count, charged GAE, or whole-pipeline S. Seq 58, for example, retains `total_gae=741.091311`, but current x86_64 counts 301,961 solver word XORs where the archived Apple record counts 300,531. The committed same-host audit made with the measuring binary reports 168/168 identical. Until this cross-host counter drift is explained or the method identity is versioned, the root-finder study's F4 stage-count comparison is historical evidence. The root-finder rows themselves replay identically.

8. Exact frontier-moving obligations

1. Register torsion-symmetrised bases and N83 dense root finding under new, immutable ECBench method IDs, with cold and five-phase online records.
2. Resolve inherited-F4 word-XOR cross-host replay drift before fitting a bound.
3. Run a native challenge at four or more sizes against the standing rho bound.
4. Repeat N83 on a Tier 1/0 isolated host, charging every cold phase and byte.
5. Pin a conversion for F6′s batched geometry before interpreting wall/GAE disagreement.

Non-claims. Public synthetic known-answer inputs only; no deployed-curve or key-recovery claim; no ECC2K-130 transfer; no new exponent; no canonical scoreboard promotion. Existing canonical graphs were checked and left unchanged. The required Conductor executable and workflow file were absent, so work stayed inside this report, benchmark copies, and benchmark index.